Project 2_Game Analysis

November 1, 2026

1 Descriptive Statistics Project

1.1 Gaming Behavior: A Comparison Between the United States and the United Kingdom

1.2 1. Project Overview

This project explores **gaming behavior among participants in the United States (US) and the United Kingdom (UK)** using descriptive statistics.
The analysis will summarize demographic characteristics and gaming habits for each country.

2 2. Data Preparation

2.0.1 Load the dataset

Import the dataset

2.0.2 Key variable

- `country` – indicates whether the participant is from the **US** or **UK**.

You will use this variable to **split or group the data** when calculating statistics.

3 3. Step 1 — Describe the Demographics of Each Country

3.1 Research Question

What are the demographic characteristics of participants in the US and UK samples?

3.1.1 Variables to analyze

- `age`
- `gender`
- `ethnicity`
- `edu_level`
- `employment`
- `marital_status`

3.1.2 Numerical variables

For numerical variables, calculate:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- Range
- IQR
- plot a histogram with a title.

And report each value in a table (or tables).

3.1.3 Categorical variables

For each categorical variable, visualize it with:

- Bar graph
- For **gender**, let's show **both countries in one bar graph**.
- For **ethnicity**, let's plot a bar graph for **each country**.
- All the others: Your choice

3.1.4 Quick analysis

What did you find here? Do you see any similarities and differences between the two groups? Explain.

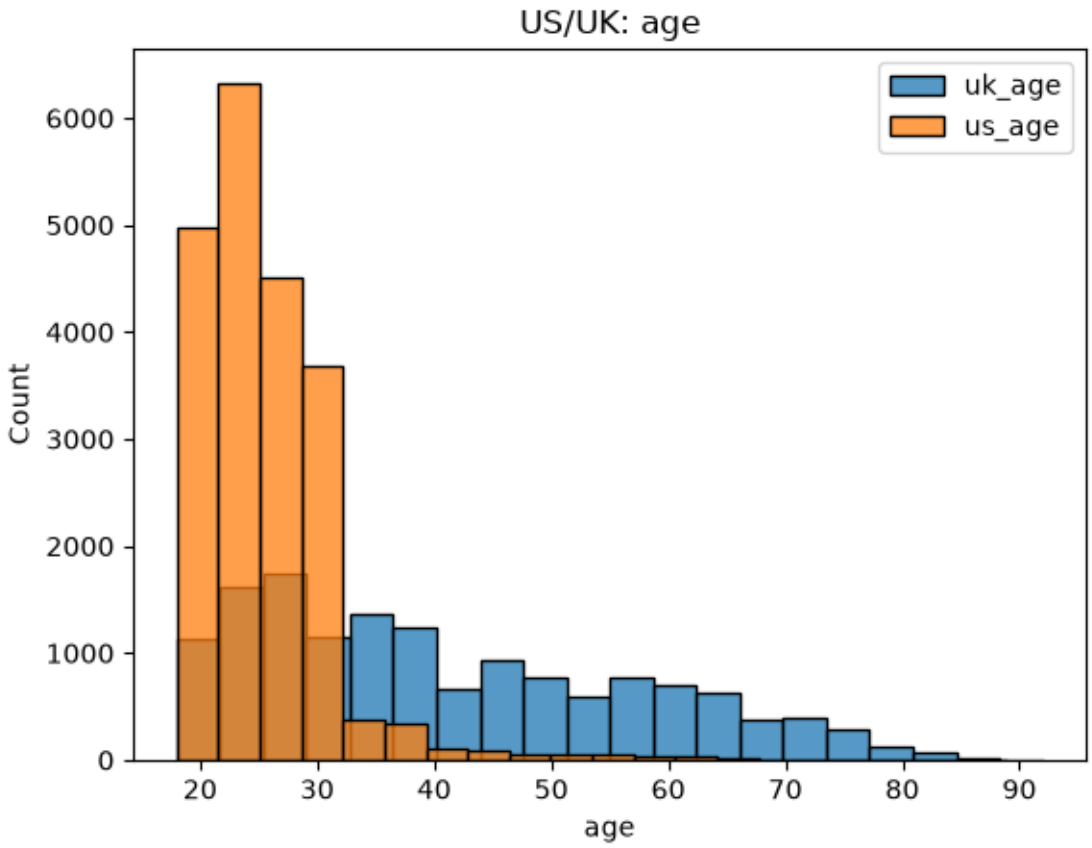
[64]:
```
#setup
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("/home/abrooks/GitHub/CS138_TMCC/bigProject2/
 ↪survey_intake_cleaned.csv")
```

```
/tmp/ipykernel_642514/3556511240.py:6: DtypeWarning: Columns
(10,38,39,41,62,63,66,78,79,80,81,82,83,84) have mixed types. Specify dtype
option on import or set low_memory=False.
  df = pd.read_csv("/home/abrooks/GitHub/CS138_TMCC/bigProject2/survey_intake_cl
eaned.csv")
```

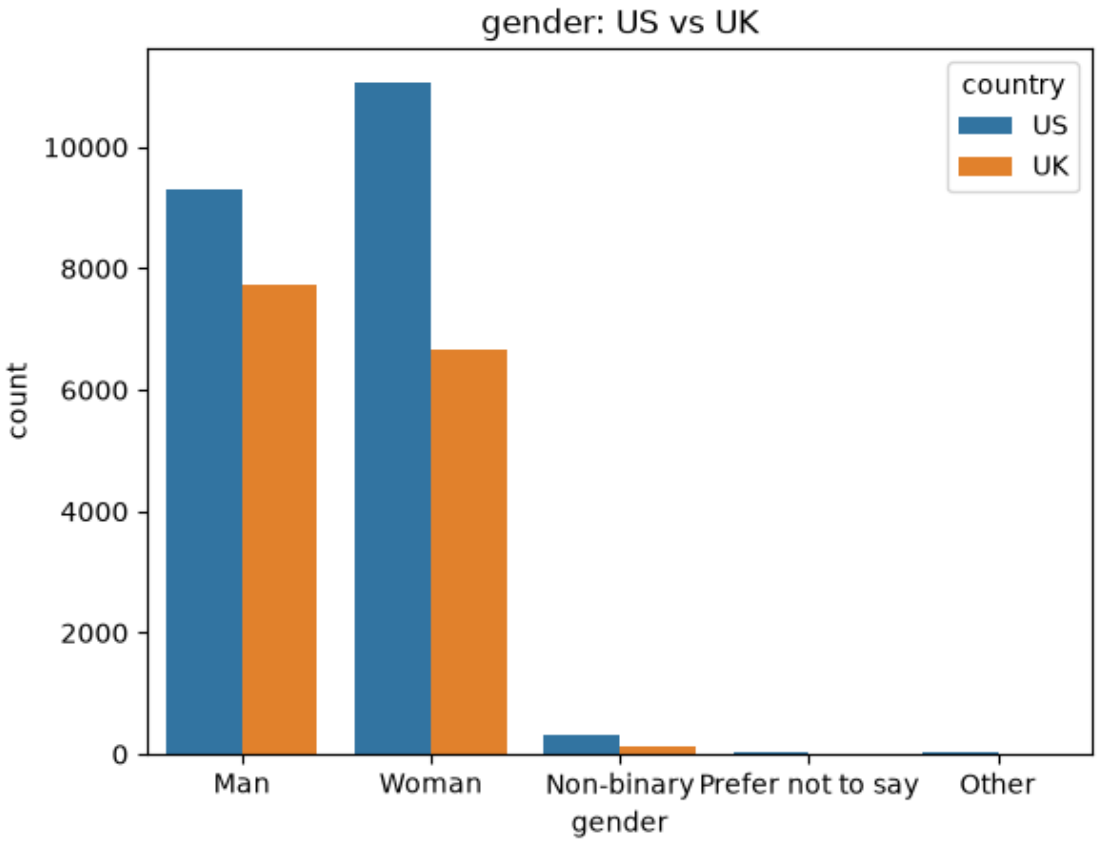
[65]:
```
df_us = df.query("country == 'US'")
df_uk = df.query("country == 'UK'")
df_both = df.query("country == 'UK' or country == 'US'")

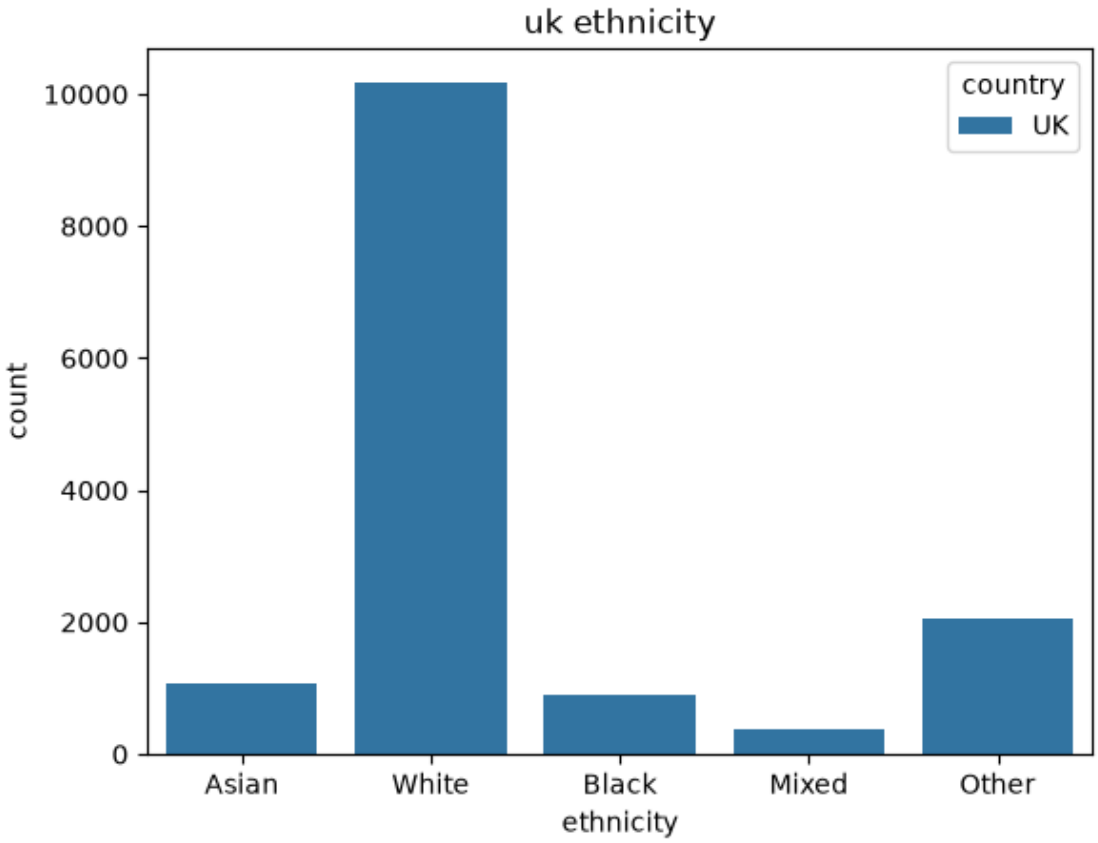
variables = ["age", "gender", "ethnicity", "edu_level", "employment",␣
 ↪"marital_status"]
sets = {"uk":df_uk,"us":df_us}
Stat_summary = pd.DataFrame(columns=['mean', 'median', 'std', 'min', 'max',␣
 ↪'range', 'IQR'])

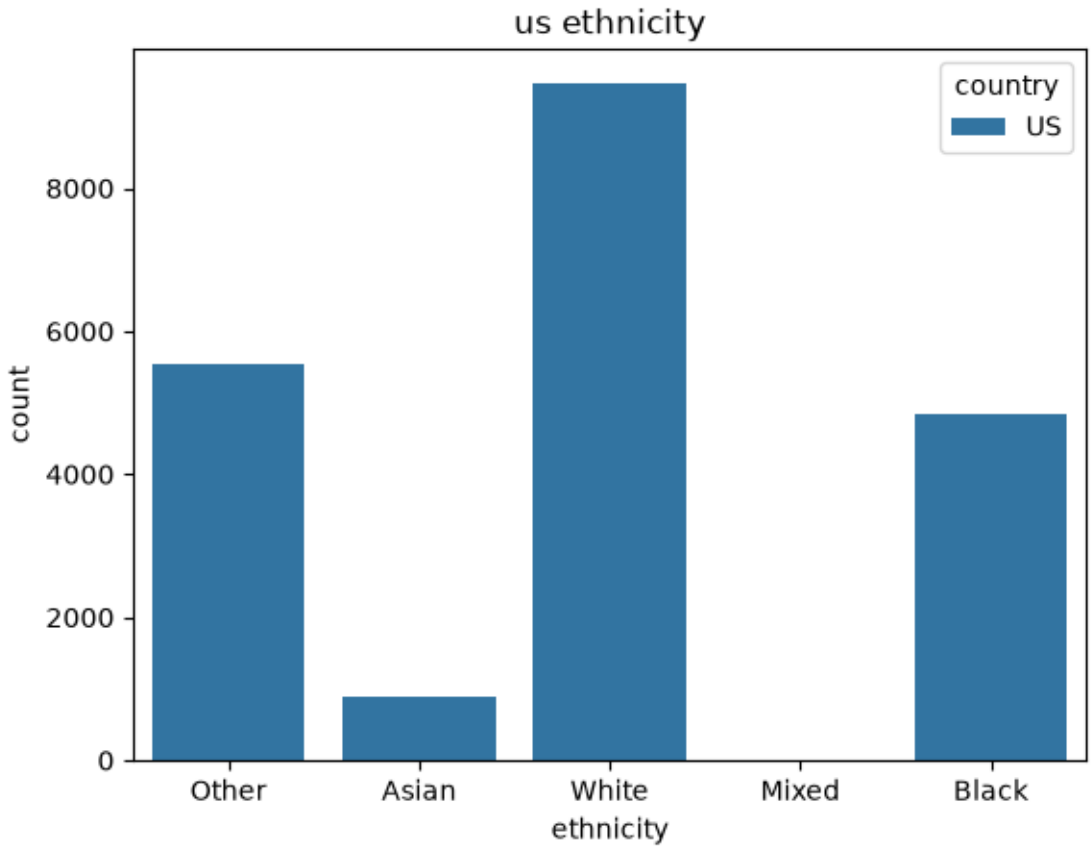
for variable in variables:
    if variable == "age":
        for name, data in sets.items():
                Stat_summary.loc[name + "_" + variable] = [round(data[variable].
 ↪mean(),2),round(data[variable].median(),2),
                    round(data[variable].std(),2),␣
 ↪min(data[variable]),max(data[variable]),
                    (max(data[variable])-min(data[variable])),(data[variable].
 ↪quantile(0.75) - data[variable].quantile(0.25))]
                sns.histplot(data[variable],label=name + "_" + variable,bins =␣
 ↪20)
                plt.title("US/UK: " + variable)
                plt.legend()
        plt.show()
        display(Stat_summary)
    elif variable == "ethnicity":
        for name, data in sets.items():
            sns.countplot(x=variable, hue='country', data=data)
            plt.title(name + " " + variable)
            plt.show()
    else:
        sns.countplot(x=variable, hue='country', data=df_both)
        plt.title(variable + ": US vs UK")
        plt.show()
```

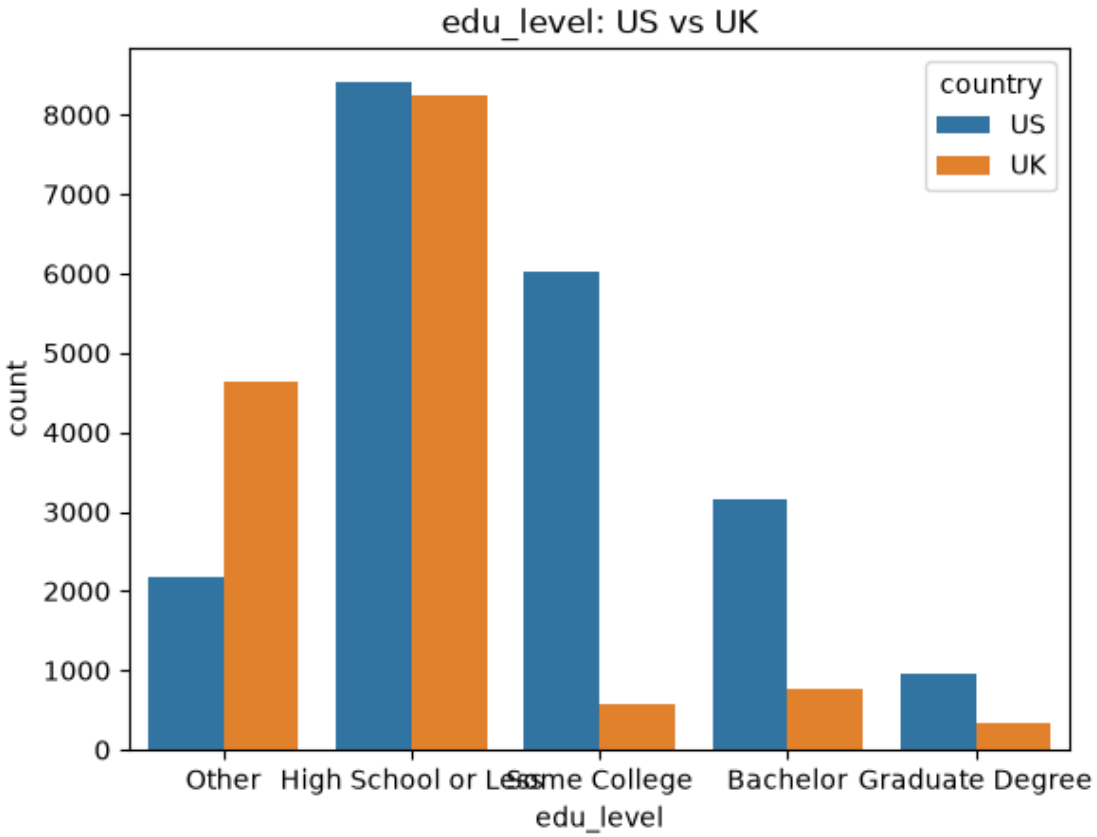


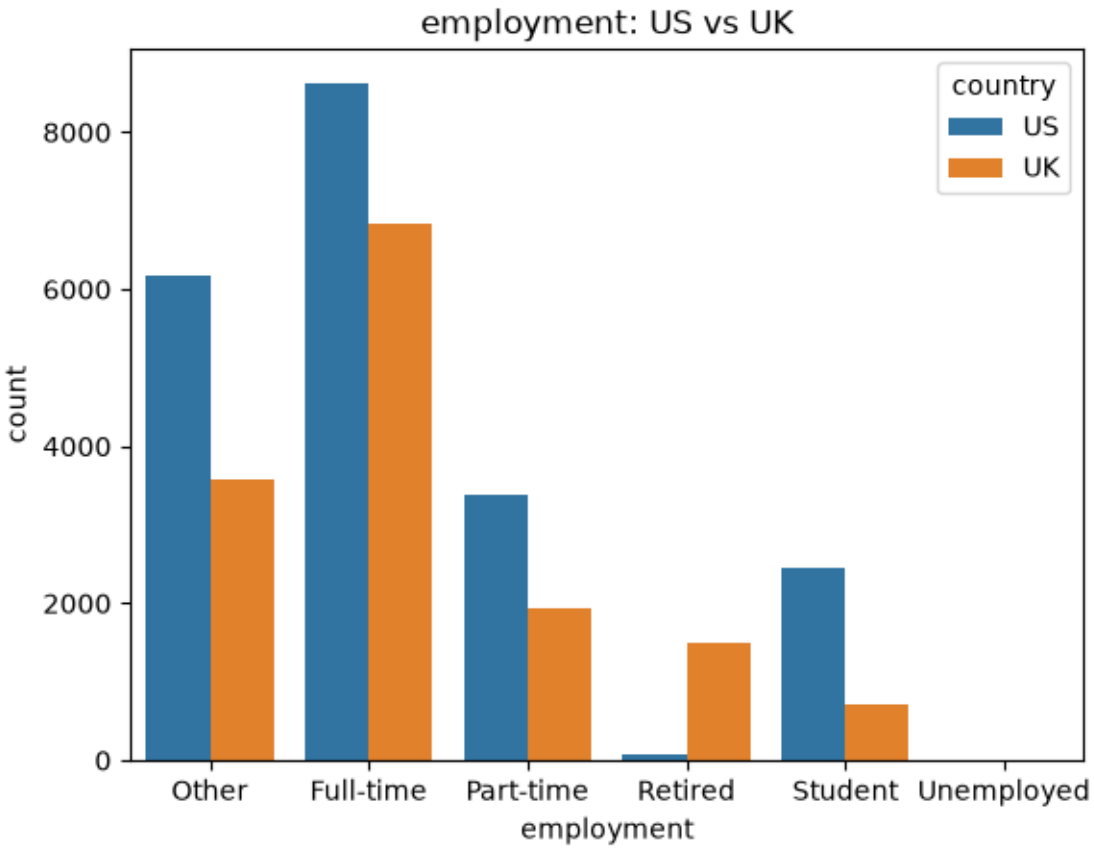
	mean	median	std	min	max	range	IQR
uk_age	40.95	37.0	16.12	18.0	92.0	74.0	26.0
us_age	25.45	25.0	5.75	18.0	89.0	71.0	6.0

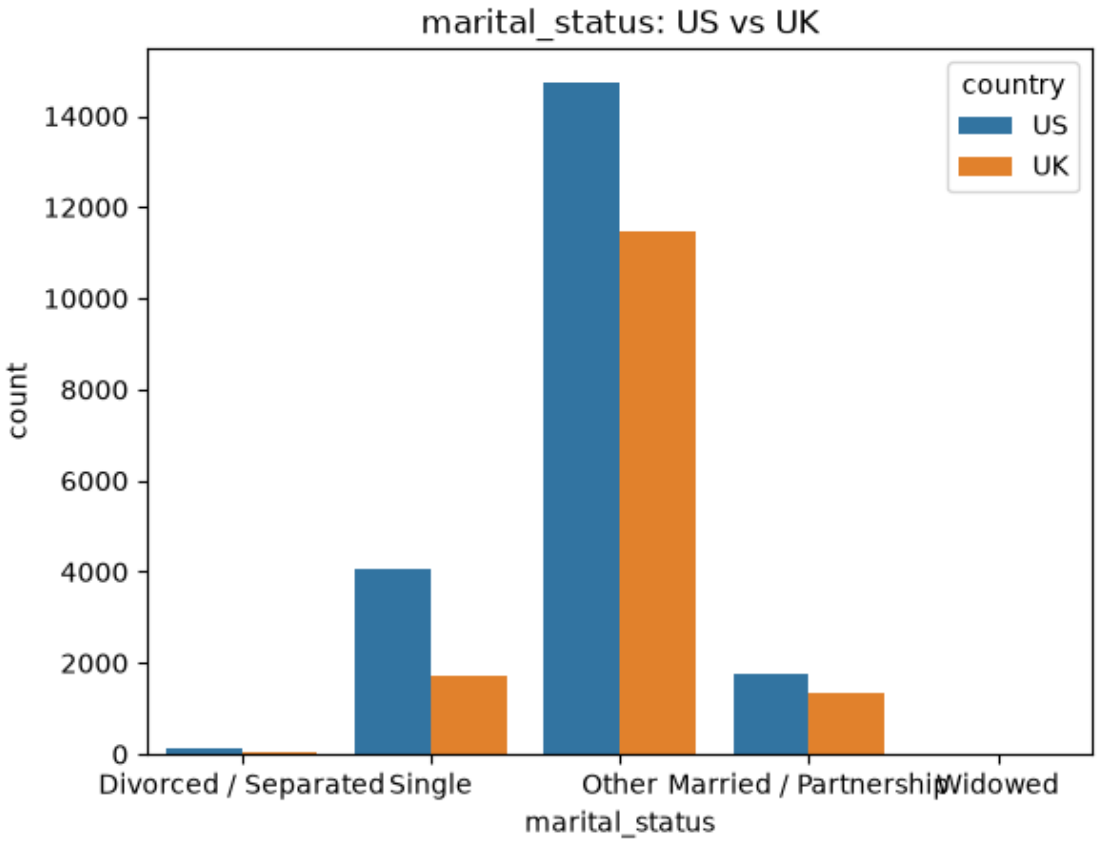

gender: US vs UK
country
US
UK
count
10000
8000
6000
4000
2000
0
Man
Woman
Non-binary
Prefer not to say
Other
gender


uk ethnicity
country
UK
10000
8000
6000
4000
2000
0
count
Asian
White
Black
Mixed
Other
ethnicity


us ethnicity
country
US
8000
6000
4000
2000
0
count
Other
Asian
White
Mixed
Black
ethnicity


edu_level: US vs UK
country
US
UK
count
8000
7000
6000
5000
4000
3000
2000
1000
0
Other
High School or Less
Some College
Bachelor
Graduate Degree
edu_level


employment: US vs UK
country
US
UK
count
8000
6000
4000
2000
0
Other
Full-time
Part-time
Retired
Student
Unemployed
employment



3.2 Quick analysis

The first thing, and perhaps the most dramatic difference between the US and UK data, is the difference in age. As seen in the first graph, US players are dramatically younger leaning compared to the much more spread out distribution of UK players. This difference can be seen even in our summary statistics, such as the mean, where US players average around 25 while UK players average around 41. Although, the mean may not be the best summary for the UK, because the older players skew the mean above the median of 37. The UK also has a much higher std of ~16 compared to the US's ~6. The range of both is very similar, with 74 and 71, with the UK taking the lead, as well as leading in IQR with 26.

Gender highlights the surprising disparity of more female samples than male in the US. Considering this data is self-reported, this may be an artifact of female players being more willing to participate in the study overall, and shouldn't be used as an indication that there are necessarily more female players than male within the total population.

Ethnicity reports white participants as the largest group in both predominantly white nations, though only the UK has a white majority, as well as a higher proportion of Black participants in the US, as one might expect.

Education is another place we see an interesting disparity, where it would seem that US players of higher education are much more likely to participate than in the UK, as early as the Some College

category, and that trend is continued all the way through graduates.

4 4. Step 2 — Compare Gaming Behavior

4.1 Research Question

How does gaming behavior differ between gamers in the US and the UK?

4.1.1 Variables

- `self_reported_weekly_play`
- `plays_steam`
- `plays_playstation`
- `plays_xbox`
- `plays_nintendo`
- `plays_ios`
- `plays_android`

4.1.2 Weekly gaming time

Calculate for each country:

- Mean weekly play time
- Median weekly play time
- Standard deviation
- Plot a histogram for each country

And report each value in a table (or tables).

4.1.3 Platform usage

- **Calculate** and find how many people use each platform (Hint: You can calculate it with `sum()` since each platform variable is `boolean`.)
- Report each platform total for each country in a **table**

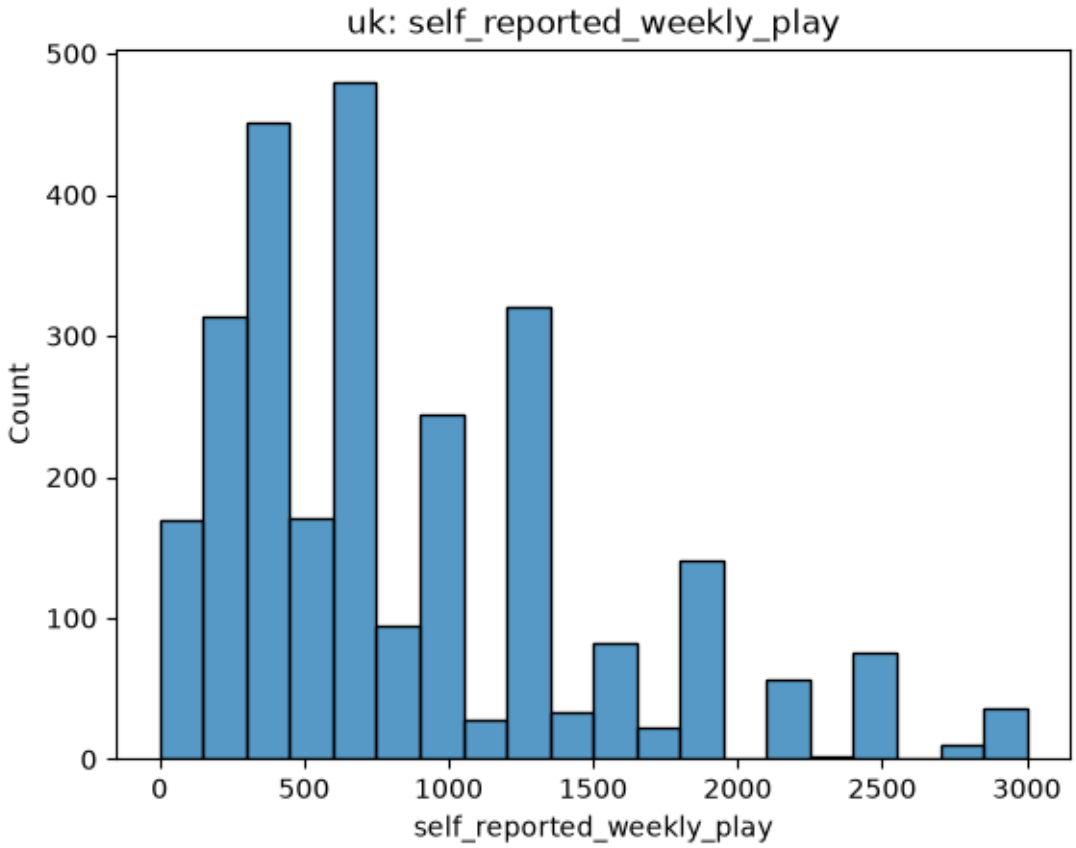
4.1.4 Quick analysis

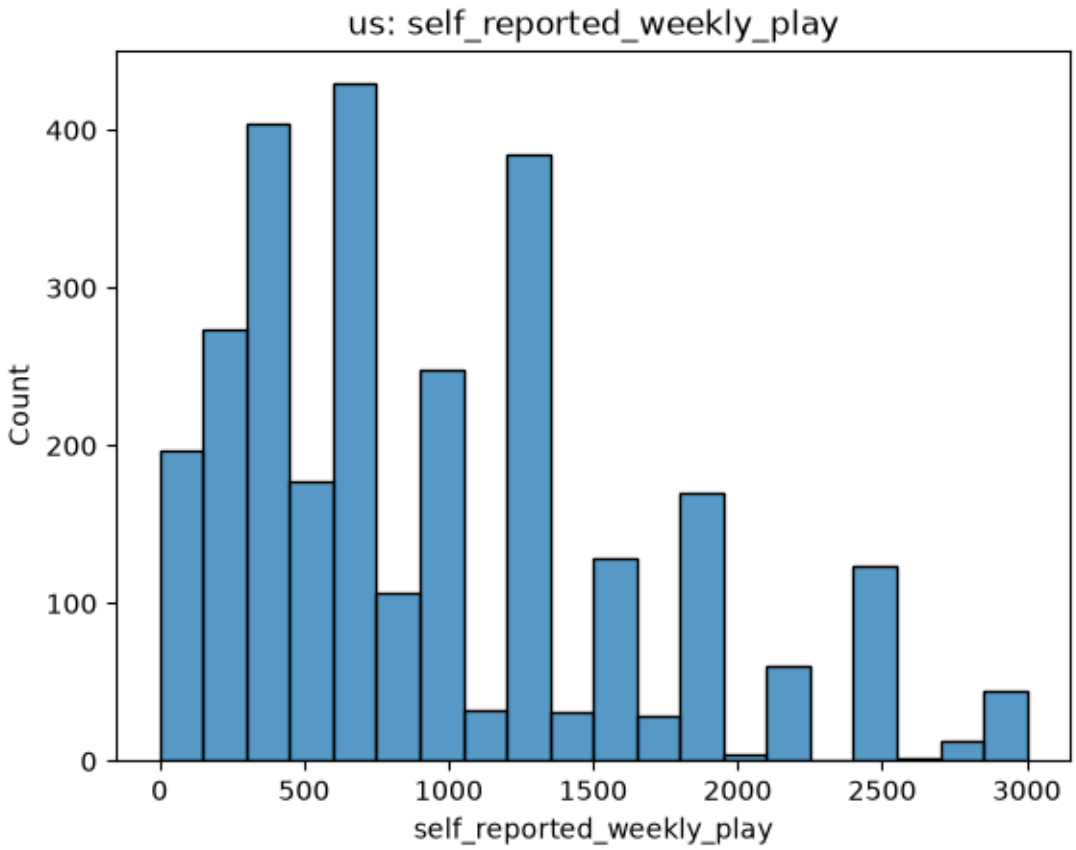
What did you find here? Did you see any similarities and differences between the two groups? Explain.

[66]:
```
variables = ["self_reported_weekly_play", "plays_steam", "plays_playstation",
    "plays_xbox", "plays_nintendo", "plays_ios", "plays_android"]

weekly_summary = pd.DataFrame(columns=['mean', 'median', 'std'])
platform_summary = pd.DataFrame(columns=['plays_steam', 'plays_playstation',␣
 ↪'plays_xbox',
                                          "plays_nintendo", "plays_ios",␣
 ↪"plays_android"])
for variable in variables:
```

```
    if variable == "self_reported_weekly_play":
        for name, data in sets.items():
            weekly_summary.loc[name + "_" + variable] = [round(data[variable].
 ↪mean(),2),
                round(data[variable].median(),2),round(data[variable].std(),2)]
            sns.histplot(data[variable],label=name + "_" + variable,bins = 20)
            plt.title(name+": " + variable)
            plt.show()
        display(weekly_summary)
    else:
        for name, data in sets.items():
            platform_summary.loc[name, variable] = data[variable].sum()
display(platform_summary)
```






```
                              mean  median     std
uk_self_reported_weekly_play  825.07   600.0  637.21
us_self_reported_weekly_play  907.03   720.0  680.06

   plays_steam plays_playstation plays_xbox plays_nintendo plays_ios  \
uk        3443              5964       3388           2706      4038
us        4392              6354       4862           3147      6386

   plays_android
uk          5098
us          3730
```

5 5. Step 3 - Conclusion

Based on your findings from Steps 1 and 2, can you draw some conclusions? Explain what the results suggest about **gaming behavior across countries** related to each country's demographic characteristics.

5.1 Conclusion

Based on these findings, it would appear that despite the demographic differences identified, the total weekly reported play has similar distributions, with the mean play in the US a bit higher with 907, compared to the UK's 825. Platforms for users show interesting results, with more respondents in the US for every platform excluding Android, which shows a much higher 5k respondents in the UK compared to the 3.7k in the US. However, the US sample is larger overall, so higher US totals are somewhat expected and don't necessarily mean US players favor those platforms more. This could suggest that the UK sample may favor the Android platform over iOS for mobile gaming, considering not only are they higher than the US in total, but compared to their own iOS of 4k, a reversal of the trend seen for the US.

6 6. Step 4 - Further questions

Based on your conclusion, what further questions would you come up with to expand this research in the future?

6.1 Further questions

The first place to start moving forward would be to scale both datasets so we could compare the differences normalized across samples. Considering the large disparity in US respondents to UK, it makes it difficult to do a proper comparison of the two. It's also worth noting that the original dataset had a small subset of participants with the country labeled "Other." Although it was small and the initial filtering removed these, it would be interesting to look closer at those samples to see how they differ from the US and UK. Beyond that, the age gap between the two samples is large enough that it could be influencing other results, and the UK's lean toward Android over iOS, the reverse of the US, is worth looking into further.

Questions moving forward:

1: How do the US and UK compare once the data is normalized for sample size, and do the platform differences still hold?

2: Does the large age difference between the US and UK samples explain some of the other differences, such as education or platform choice?

3: Why does the UK sample favor Android over iOS for mobile gaming when the US shows the opposite trend?

7 7. Step 5 - Reflection

Write a short reflection (aim for at least 3 sentences in paragraphs) on your experience completing this assignment. You may write about what was challenging, mental breakthroughs, what you learned, etc.

7.1 Reflection

I really enjoyed this one. I like how it was more open-ended and left room for more analysis. I also enjoyed creating the structure of part one that I could mostly recycle in part two. Using filtering and loops to cut down on lines, as well as thinking through the dataset logically, is always fun.

That being said, the loops I created still use hardcoded variables, so they couldn't be used in other contexts, but with some refactoring I think they could be added to a personal library to be used on other analysis. One possibility is removing the columns you're not using for the analysis and ordering them in a way that makes loops easier. This is what I do a lot in R (a leftover from Excel being my primary tool), and it would significantly improve the simplicity of looping over columns.